

To obtain a subjective judgement whether a book is STEM or non-STEM, the annotator should answer the following questions:

- Does the publisher classify the book under Science, Technology, Engineering, or Mathematics?
- Do the title, book description, and publisher summary/categories indicate that the primary purpose is teaching, explaining, or exploring a STEM subject?

Are the dominant topics STEM according to the annotation guidelines? The guidelines to manually define exactly what counts as STEM are:

Science: Biology, Physics, Chemistry, Astronomy, Earth Science

Technology: Programming, AI, Robotics, Computing

Engineering: Mechanical, Electrical, Civil, Aerospace

Mathematics: Algebra, Geometry, Statistics, Calculus

Non-STEM: History, Romance, Fiction, Biography, Religion, Politics, Art. Science fiction works are classified under Non-STEM except in instances where scientific or technological elements serve as central narrative drivers rather than merely decorative backdrop.

If at least two criteria are satisfied, classify the book as STEM.

- * Annotators do not have to be domain experts (e.g., librarians, educators, or STEM faculty) for this type of annotation. In information retrieval, NLP, and recommender systems, it is common for trained research assistants or graduate students to perform annotation, provided there is a well-defined annotation protocol.
- * Use the Fleiss' k formula to determine the degree of agreement between the annotators. If Cohen's κ exceeds approximately 0.80, then the annotation process is regarded as highly reliable, indicating that independent annotators can consistently apply the annotation guidelines.